

1. A manager has prepared a table that shows the cost of performing each of four jobs by each of four employees as shown below.

		Employee			
		A	B	C	D
Job	1	\$15	20	18	24
	2	12	17	16	15
	3	14	15	19	17
	4	11	14	12	13

Required

- a. Work out the optimal assignment plan for the problem

Solution. Q.1

	Employee			
	A	B	C	D
1	15	20	18	24
2	12	17	16	15
Job 3	14	15	19	17
4	11	14	12	13

To do this follow the following step

Step 1 Row and Column min Reduction / operation

1. - Subtract the minimum value of each row from the entries of that row
2. Subtract the minimum value of each Column from the entries of that Column.

	Employee				
	A	B	C	D	Row min
1	15	20	18	24	15
2	12	17	16	15	12
Job 3	14	15	19	17	14
4	11	14	12	13	11

② Row operation

	A	B	C	D
1	0	5	3	9
2	0	5	4	3
Job 3	0	1	5	3
4	0	3	1	2

③ Column min D 1 1 2

④ Column operation

	A	B	C	D
1	0	4	2	7
2	0	4	3	1
Job 3	0	0	4	1
4	0	2	0	0

⑤ Cover all zeros by rows and by Column.

	A	B	C	D
1	0	4	2	7
2	0	4	3	1
Job 3	0	0	4	1
4	0	2	0	0

The solution has no feasible b/c ~~row~~ No of row and Circled are not equal

⑥ select minimum undelated no then that select min. no, add in corner and subtract from under

	A	B	C	D
1	0	4	1	6
2	0	4	2	0
3	0	0	3	0
4	1	3	0	0

	A	B	C	D
1	0	4	1	6
2	X	4	2	0
3	X	0	3	X
4	1	3	0	X

job 1 → A → 15
 job 2 → D → 15
 job 3 → B → 17
 job 4 → C → 12

∴ The soln has a feasible

Total cost = 59 \$

2. Harvey Company has contracted to provide topsoil for three residential housing developments. Topsoil can be supplied from three different farms as follows:

From	Weekly capacity (cubic yards)
A	100
B	200
C	200

A demand for the topsoil generated by the construction project is:

<i>Project</i>	<i>Weekly demand (cubic yards)</i>
# 1	50
# 2	150
# 3	300

The manager of Harvey Company has estimated the cost per cubic yard to ship over each route:

	<u><i>Cost per cubic yard</i></u>					
<i>From</i>	<u><i>Project # 1</i></u>		<u><i>Project # 2</i></u>		<u><i>Project # 3</i></u>	
Form A	Birr	4	Birr	2	Birr	8
Form B		5		1		9
Form C		7		6		3

Required:

- Formulate transportation table/matrix
- Determine the optimal shipment routes that minimize the total transportation cost satisfying the supply and demand conditions.(use NWCM, LCM and VAM)

Solution. Q.2.A

	Pr#1	Pr#2	Pr#3	Weekly Capacity
A	4	2	8	100
B	5	1	9	200
C	7	6	3	200
dd	50	150	300	500

Solution. Q.2.B. By NWCM

	Pr#1	Pr#2	Pr#3	Supply
A	<u>50</u> 4	<u>50</u> 2	8	100 50
B	5	<u>100</u> 1	<u>100</u> 9	200 100
C	7	6	<u>200</u> 3	200 0
dd	50 0	150 0	300 200	

	Pr#1	Pr#2	Pr#3	Supply
A	<u>50</u> 4	<u>50</u> 2	8	100
B	5	<u>100</u> 1	<u>100</u> 9	200
C	7	6	<u>200</u> 3	200
dd	50	150	300	

Total cost = $50 \times 4 + 50 \times 2 + 100 \times 1 + 100 \times 9 + 200 \times 3 = \text{Br. } \underline{1900}$

Solution. Q.2.B. By LCM

	Pr#1	Pr#2	Pr#3	Supply
A	<u>50</u> 4	2	<u>50</u> 8	100 50
B	5	<u>150</u> 1	<u>50</u> 9	200 50
C	7	6	<u>200</u> 3	200 0
dd	50 0	150 0	300 100	

	Pr#1	Pr#2	Pr#3	Supply
A	<u>50</u> 4	2	<u>50</u> 8	100
B	5	<u>150</u> 1	<u>50</u> 9	200
C	7	6	<u>200</u> 3	200
dd	50	150	300	

Total cost = $50 \times 4 + 150 \times 1 + 50 \times 8 + 50 \times 9 + 200 \times 3 = \text{Br. } \underline{1800}$

Solution 2.B. By VAM Method.

	Pr#1	Pr#2	Pr#3	Supply	P _{C1}	P _{C2}	P _{C3}
A	<u>50</u> 4	2	<u>50</u> 8	100 50	2	2	(4)
B	5	<u>150</u> 1	<u>50</u> 9	200 50	4	(4)	4
C	7	6	<u>200</u> 3	200	3	-	-
dd	50	150	300				

P _{C1}	1	1	
P _{C2}	1		(5)
P _{C3}	1	-	1

	Pr#1	Pr#2	Pr#3	Supply
A	<u>50</u> 4	2	<u>50</u> 8	100
B	5	<u>150</u> 1	<u>50</u> 9	200
C	7	6	<u>200</u> 3	200
dd	50	150	300	

Total Cost = $50 \times 4 + 150 \times 1 + 50 \times 8 + 50 \times 9 + 200 \times 3 = \text{Br. } 1800$

3. A firm produces three products A, B, and C. These products pass through two machines M and Q. Each machine has the availability of 30 and 60 hours, respectively. Each A requires 2 hrs of machine M and 1 hour of machine Q. Each B requires 2 hrs of machine M and 3 hrs of Q. Each C requires 3 hrs of M and 5 hrs of Q. The firm receives a net contribution of \$5, \$4, & \$3 per unit of A, B, & C produced, respectively

Required: Solve using Simplex method

Solution. Q.3

	A	B	C	Available hour
M	2	2	3	30
Q	1	3	5	60
Net Cont	5	4	3	

Step 1 formulate model

objective function
 $Z_{\max} = 5x_1 + 4x_2 + 3x_3$

Sub to

$$2x_1 + 2x_2 + 3x_3 \leq 30$$

$$x_1 + 3x_2 + 5x_3 \leq 60$$

} Constraint

x_1, x_2 and $x_3 \geq 0$ (non-negativity)

Step 2 Write the LPM in a standard form.

$$Z_{\max} = 5x_1 + 4x_2 + 3x_3 + 0s_1 + 0s_2$$

Sub to

$$2x_1 + 2x_2 + 3x_3 + s_1 = 30$$

$$x_1 + 3x_2 + 5x_3 + s_2 = 60$$

x_1, x_2, x_3, s_1 and $s_2 \geq 0$ (Non-negativity)

Step 3: Construct the initial Simplex tableau.

BV	C_j	5	4	3	0	0	BS/RHS	Ratio
		x_1	x_2	x_3	s_1	s_2		
s_1	0	2	2	3	1	0	30	$30/2 = 15$
s_2	0	1	3	5	0	1	60	$60/1 = 60$
Z_j		0	0	0	0	0	0	
$C_j - Z_j$		5	4	3	0	0		

→ Pivot Column.

Entering Variable = x_1 , leaving Variable = s_1 , Key Element = 2

$$\frac{1}{2} R_1 \rightarrow R_1 \rightarrow \frac{1}{2} (2 \ 2 \ 3 \ 1 \ 0 \ 30)$$

$$\text{New } R_1 \rightarrow 1 \ 1 \ 3/2 \ 1/2 \ 0 \ 15$$

$$- \text{New } R_1 + R_2 \rightarrow R_2$$

$$-1 \ -1 \ -3/2 \ -1/2 \ 0 \ -15$$

$$+1 \ +3 \ 5 \ 0 \ 1 \ 60$$

$$0 \ 2 \ 7/2 \ -1/2 \ 1 \ 45 \rightarrow \text{New } R_2$$

BV	C_j	5	4	3	0	0	BS/RHS
		x_1	x_2	x_3	s_1	s_2	
x_1	5	1	1	$3/2$	$1/2$	0	15
s_2	0	0	2	$7/2$	$-1/2$	1	45
Z_j		5	5	$15/2$	$5/2$	0	75
$C_j - Z_j$		0	-1	$-9/2$	$-5/2$	0	

The solution is optimal b/c in $C_j - Z_j$ row all value are zero and negative number $Z_{\max}$: $C_j - Z_j \leq 0$.

$$x_1 = \underline{15}, x_2 = \underline{0}, s_2 = \underline{45}, Z_{\max} = \underline{75}$$

4. Solve the following model using Graphical method

$$\text{Max } Z = 4X_1 + 3X_2$$

St:

$$20X_1 + 10X_2 \leq 230 \text{ } \textit{Orange Constraint}$$

$$5X_1 + 15X_2 \leq 120 \text{ } \textit{Grape Constraint}$$

$$\text{And } X_1, X_2 \geq 0 \text{ } \textit{Non-negativity constraint}$$

Solution Q. 4

$$Z_{\max} = 4x_1 + 3x_2$$

Subto

$$20x_1 + 10x_2 \leq 230$$

$$5x_1 + 15x_2 \leq 120$$

$$x_1 \text{ and } x_2 \geq 0$$

Step 1 Convert constraint inequalities to equalities

$$20x_1 + 10x_2 = 230$$

$$5x_1 + 15x_2 = 120$$

Step 2 find intercept

$$20x_1 + 10x_2 = 230$$

$$x_1 = 0, x_2 = 23 \rightarrow (0, 23)$$

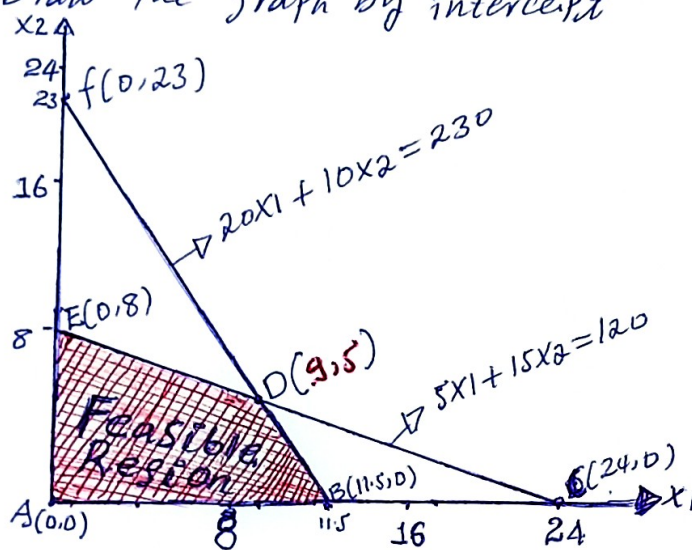
$$x_2 = 0, x_1 = 11.5 \rightarrow (11.5, 0)$$

$$5x_1 + 15x_2 = 120$$

$$x_1 = 0, x_2 = 8 \rightarrow (0, 8)$$

$$x_2 = 0, x_1 = 24 \rightarrow (24, 0)$$

Step 3 Draw the graph by intercept



$$\begin{array}{rcl} 20x_1 + 10x_2 & = & 230 \\ -20x_1 - 60x_2 & = & -480 \\ \hline -50x_2 & = & -250 \\ \hline x_2 & = & 5 \end{array}$$

$$\underline{x_2 = 5}$$

$$5x_1 + 15x_2 = 120$$

$$5x_1 + 15(5) = 120$$

$$5x_1 + 75 = 120$$

$$5x_1 = 120 - 75$$

$$x_1 = \frac{45}{5}$$

$$\underline{x_1 = 9}$$

Corner Point

$$A(0,0)$$

$$B(11.5,0)$$

$$D(9,5)$$

$$E(0,8)$$

Optimal Soln

$$4(0) + 3(0) = \underline{0}$$

$$4(11.5) + 3(0) = \underline{46}$$

$$4(9) + 3(5) = \underline{51}$$

$$4(0) + 3(8) = \underline{24}$$

There for $x_1 = \underline{9}$ and $x_2 = \underline{5}$ to get maximum Profit of 51